Richard Gonzalez

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SUMMARY

Software engineer with 3+ years of experience designing and developing applications using Java, Spring Boot, Angular, and PostgreSQL. Collaborates directly with business users to translate requirements into features, optimize performance, and improve software quality. Independently builds and deploys AI-integrated applications on AWS using Terraform and CI/CD.

EXPERIENCE

Software Engineer Jan 2023 – May 2026

General Motors *Austin, TX*

Developed and maintained vehicle-configuration applications serving ~300 business users across 40+ vehicle programs, along with a supplier-warranty claims platform.

- Engineered an automated vehicle-option rule management feature using Angular, Spring Boot, and PostgreSQL, lowering manual updates by 50% and preventing invalid data from reaching downstream systems.
- Optimized PostgreSQL query performance using JMeter and EXPLAIN to identify and fix an unbounded subquery, cutting vehicle-catalog load times from ~20s to 2s (truck catalogs from ~2 min to 20s) across 1,000+ catalogs.
- Migrated a vehicle-program application from Pivotal Cloud Foundry to on-premises Kubernetes ahead of a deadline carrying $1M+ in penalties, building deployment manifests and a CI/CD pipeline reused by three other teams.
- Modernized a security-flagged vehicle-configuration application by independently upgrading Angular 15 to 18 and Spring Boot 2.7 to 3.4, resolving PrimeNG and Spring Security compatibility issues to restore security support.
- Automated regression testing by introducing the frontend's first Jest and Playwright suites (0% to 50% coverage) and increasing JUnit coverage from ~20% to 90% across seven microservices, shifting defect detection from UAT into CI.

PROJECTS

Vehicle Catalog | catalog.rgonz.dev | github.com/rgonzal92/vehicle-catalog 2026

Java, Spring Boot, Spring AI, Angular, PostgreSQL, Redis, AWS, Terraform

Solo-built, AWS-hosted catalog management platform with demo access, approval workflows, versioning, and AI-assisted rule drafting. Architecture decisions documented in GitHub.

- Architected the full-stack platform, including a 48,000-cell autosaving grid, rule validation, approvals, and three-way merge, reducing the largest save operation by 89% (517 ms to 57 ms).
- Integrated OpenAI through Spring AI to power rule drafting, tool-calling analysis, and RAG document search with pgvector, enforcing Java-side output validation and an atomic $1/day AI spending limit.
- Codified AWS infrastructure and deployments using Terraform and GitHub Actions, implementing keyless OIDC authentication, health-check-based rollback, and provisioning across EC2, CloudFront, S3, Cognito, SQS, and ECR.
- Built an SQS-backed worker with dead-letter handling for exports, embeddings, and AI summaries, alongside Redis caching with PostgreSQL fallback.
- Instrumented the backend with CloudWatch metrics, dashboards, and alarms, eliminating deployment-related 502 errors through request-level testing and release improvements.

EDUCATION

The University of Texas at San Antonio San Antonio, TX

Bachelor of Science in Computer Science *May 2022*

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript, SQL
Frontend: Angular, RxJS, PrimeNG, Angular Material, AG Grid, Tailwind CSS
Backend & Data: Spring Boot, Spring Security, Spring AI, Spring Data JPA, REST APIs, PostgreSQL, pgvector, Redis, Flyway
Cloud & DevOps: AWS (EC2, S3, CloudFront, SQS, Cognito, IAM), Terraform, Docker, Kubernetes, Git, GitHub Actions (CI/CD)
Testing & Observability: JUnit 5, Testcontainers, Jest, Playwright, JMeter, CloudWatch, OpenTelemetry, Grafana